

MONITORING ALGAE BLOOMS IN LOUGH NEAGH, UNITED KINGDOM USING SENTINEL-1 STATISTICAL MODELLING AND SENTINEL-2 MULTIMODAL FUSION

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ABSTRACT

Lough Neagh, the United Kingdom’s largest freshwater lake, has experienced severe eutrophication, leading to persistent harmful algal blooms (HABs) that compromise drinking water quality and ecosystem health. Conventional monitoring faces two critical limitations: in-situ sampling cannot capture spatial variability across the 383 km² lake, and optical satellite observations are frequently obscured by persistent cloud cover. To address these challenges, this work presents a multimodal framework that integrates Sentinel-1 Synthetic Aperture Radar (SAR) with Sentinel-2 optical imagery for all-weather HAB monitoring. Statistical analysis of VV polarisation backscatter reveals a consistent seasonal cycle with elevated values during bloom events relative to clean-water conditions, validating VV as a reliable biomass proxy. Building on this, we develop a multimodal deep learning architecture that fuses optical imagery, SAR intensity, and SAR-derived statistical features for region-level bloom classification. The proposed multimodal model demonstrates improvements of up to 125.8% in overall mIoU and 40.4% in macro-F1 score relative to representative baseline methods.

Index Terms— SAR, Optical, multimodal fusion, Harmful Algal Blooms, Lough Neagh.

1. INTRODUCTION

Harmful algal blooms (HABs) represent a critical global environmental crisis, affecting 11.7% of worldwide lake area and threatening aquatic ecosystems, public health, and economic stability [1]. Anthropogenic eutrophication has driven unprecedented bloom expansion over the past four decades, causing severe ecological consequences including hypoxic conditions, mass fish mortality, and trophic disruption [2]. HABs have caused significant ecological and economic impacts globally, motivating the development of

scalable monitoring solutions. Events in Lough Neagh (LN) were exceptional in both extent and duration, impacting the 383 km² lake system, including fisheries, recreation, and downstream coastal areas, with potential for extreme public health risks [3]. Invasive zebra mussels may further promote harmful cyanobacteria growth in LN [3, 4]. This ecological crisis has become politically contentious in Northern Ireland, where ethno-national interests have overshadowed environmental cooperation [5]. Beyond ecological impacts, algal blooms generate taste-and-odour compounds undermining drinking water quality and trust, highlighting the need for early-warning systems [6]. Remote sensing offers a scalable alternative to in-situ methods [7, 8], enabling automated analysis of large water bodies without physical field access.

Limitations of Conventional Monitoring: Field-based monitoring is inherently time-intensive, spatially constrained, and requires substantial labor and cost [9]. Furthermore, sparse point-based sampling fails to resolve the high spatial variability of patchy algal blooms, often leading to significant underestimations of total biomass [10]. Field campaigns are also vulnerable to adverse weather and severe bloom events, which can restrict safe access to sampling sites [11].

Constraints of Optical Sensors: Optical observations are strongly limited by cloud cover and atmospheric conditions, rendering sensors unable to provide clear imagery during uncertain weather or nighttime [10, 11].

Observational Challenges in LN: These limitations converge in LN, where persistent cloud cover compounds optical sensor constraints while dense surface blooms frequently prevent boats from accessing sampling sites. This creates a critical monitoring gap requiring an alternative approach.

To address these constraints, this work investigates a multimodal fusion framework integrating Sentinel-1 Synthetic Aperture Radar (SAR) with Sentinel-2 optical imagery for monitoring LN considering optical imagery provides a rich feature representation, while as an *active* microwave system, SAR operates day and night to monitor algal blooms in the LN, regardless of cloud cover, ensuring temporal continuity. The main contributions of this paper are:

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- **Statistical analysis framework:** We implement a statistical analysis of radar backscatter signatures (σ^0) to identify bloom occurrences. This approach is prioritised over machine learning classification to address the critical lack of region-specific labelled datasets (ground truth) for LN, avoiding the inaccuracies inherent in transferring models across distinct ecological zones.
- **Feature-based multimodal monitoring:** We demonstrate the viability of a multimodal framework that combines Sentinel-1 SAR and Sentinel-2 RGB imagery, leveraging the complementary strengths of each modality to improve monitoring robustness in a persistently cloudy region.
- **Case study:** We establish Lough Neagh as a critical case study for remote sensing, analysing a high-latitude, hypertrophic system currently experiencing a “perfect storm” of ecological decline driven by eutrophication, climate change, and invasive species.

2. METHODOLOGY

2.1. Data acquisition and preprocessing

The study focuses on LN, located in Northern Ireland, centred approximately at 54.62° N, 6.42° W. The lake’s spatial extent covers the area between latitudes 54.49° N and 54.76° N, and longitudes 6.23° W and 6.61° W (Fig. 5a). This study utilised dual-polarimetric (VV and VH) Sentinel-1 SAR data in Interferometric Wide swath mode, Ground Range Detected format, and Sentinel-2 multispectral optical imagery, both at 10 m spatial resolution and acquired from the Copernicus Open Access Hub [12], spanning the period from January 2023 to December 2025. To ensure radiometric consistency and data quality, a standard preprocessing pipeline was implemented in the SNAP toolbox. This process began with the application of precise orbit files, followed by radiometric calibration to convert digital pixel values into radar backscatter coefficients (σ^0). Subsequently, spatial speckle filtering was applied to minimise granular noise while preserving structural edges, followed by terrain correction to rectify geometric distortions. To reduce acquisition-geometry-induced variability in Sentinel-1 backscatter, we restricted the SAR time series to scenes acquired with the same orbit pass (ascending/descending) and the same relative orbit number, consistent with Tang et al. [13], who used only orbit-ascending Sentinel-1 acquisitions to ensure consistent imaging geometry, and the same relative orbit number to reduce the incidence angle effect as reported in [14]. Ground truth labels were generated using a semi-automated workflow. High density blooms were identified using Floating Algae Index (FAI), while medium density blooms were derived from chlorophyll concentration. Areas with elevated turbidity, that were not classified as high or medium, were used to represent low density blooms. The resulting labels were reviewed and manually refined to produce the final ground truth annotations.

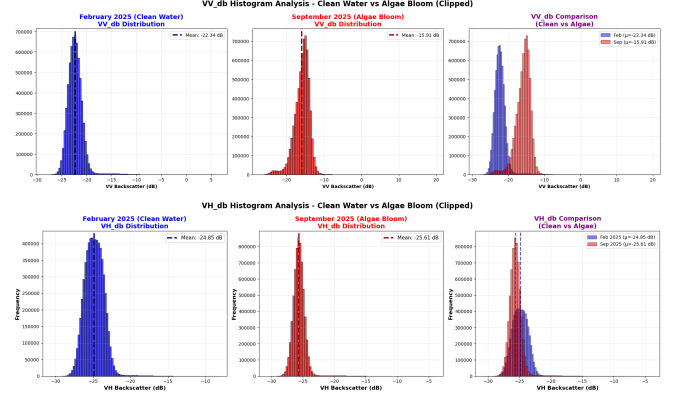


Fig. 1: Elevated whole-lake VV (top) and VH (bottom) backscatter during blooms.

Finally, all resulting imagery, SAR and optical, were co-registered to a common grid and clipped to the lake boundary to guarantee spatial comparability across the temporal series. To capture spatial variability, each Sentinel-1 and corresponding Sentinel-2 image were partitioned into non-overlapping 128×128 -pixel tiles. Given the inherent speckle noise and pixel-level instability in SAR data, we computed the linear mean VV backscatter (dB) for each tile, using these tile-averaged values as localized analysis units. This spatial aggregation approach is preferable to a single lake-wide statistic, which would obscure critical spatial patterns across the $\sim 383 \text{ km}^2$ water body.

2.2. SAR-based statistical analysis

We hypothesise that clean, winter-time water exhibits low, unimodal VV backscatter. In contrast, bloom-dominated scenes display elevated, multi-modal distributions due to mixed surface classes. This pipeline relies on radar backscattering physics: clean, calm water acts as a specular reflector, directing energy away from the sensor (low backscatter). In contrast, floating algal mats increase surface roughness and volume scattering, providing higher backscatter return [14].

We ground-truthed SAR backscatter patterns against optical RGB imagery from dates with known bloom conditions. February 2025 was selected as a bloom-free reference month because cold winter temperatures prevent algal growth, while September 2025 represented bloom conditions. To validate whether SAR backscatter reliably tracks these seasonal bloom dynamics, we compared pixel-level intensity distributions between these contrasting periods. The log-domain backscatter difference was computed as:

$$\Delta\sigma_{VV,dB}^0 = \sigma_{VV,dB}^0(\text{infested}) - \sigma_{VV,dB}^0(\text{clean}). \quad (1)$$

Histograms of $\Delta\sigma_{VV,dB}^0$ were analysed to determine if the difference indicated increased roughness (positive shift) as shown in Fig. 1.

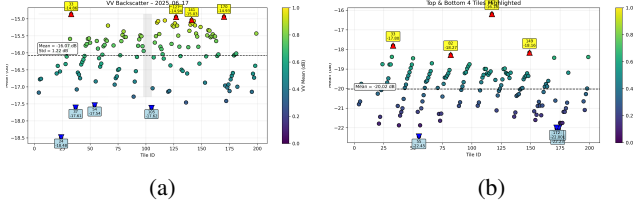


Fig. 2: Effect of wind on VV backscatter. (a) 17 Jun 2025, wind 3.20 m/s, mean VV ≈ -16.07 dB; (b) 18 Jun 2025, wind 0.865 m/s, mean VV ≈ -20.02 dB.

Analysis revealed a stark contrast between polarisations. VV backscatter increased from -22.34 dB in February to -15.91 dB in September, a shift of **+6.43 dB**. In contrast, VH showed negligible variation (-0.76 dB). This confirms that algal accumulation primarily affects the VV scattering, consistent with the VV-based approach of Wang et al. [14]. To minimise this effect, SAR images were temporally filtered to retain only acquisitions meeting the calm wind conditions described in subsection 3.1.

A key challenge in SAR-based water monitoring is the confounding effect of wind-induced surface roughness. We analysed two June 2025 acquisitions that are only one day apart. Wind conditions were characterised using ERA5 re-analysis data accessed via Google Earth Engine [15]: the 17 June acquisition experienced lake-averaged wind speeds of 3.20 m/s, while the 18 June acquisition had calmer conditions with tile-averaged wind speeds of 0.87 m/s.

Tile-wise VV backscatter for the Sentinel-1 acquisitions is shown in Fig. 2. Points denote mean VV (dB) per 128×128 non-overlapping tiles. The five highest (red triangles) and lowest (blue inverted triangles) VV tiles are highlighted to compare bloom-affected and clean-water conditions. On 17 June, the mean lake-wide VV backscatter was approximately -16.07 dB, whereas on 18 June it decreased to approximately -20.02 dB. Despite the minimal temporal separation between acquisitions, scenes acquired under higher wind speeds exhibited elevated mean VV backscatter and reduced spatial contrast, whereas low-wind scenes retained more distinct structural features consistent with bloom patches. To characterise the backscatter distribution values, Gaussian distribution was fit to the histograms, and the resulting mean (μ) and standard deviation (σ) were extracted. Specifically, a multi-component Gaussian model was fitted to each histogram:

$$f(x) = \sum_i a_i \exp \left[- \left(\frac{x - b_i}{c_i} \right)^2 \right], \quad (2)$$

where a_i , b_i , and c_i denote the amplitude, mean, and width of the i -th Gaussian component, respectively.

2.3. Multimodal fusion with SAR and optical

While SAR statistical analysis can provide an understanding of the HAB infestation, the data is often noisy, which can

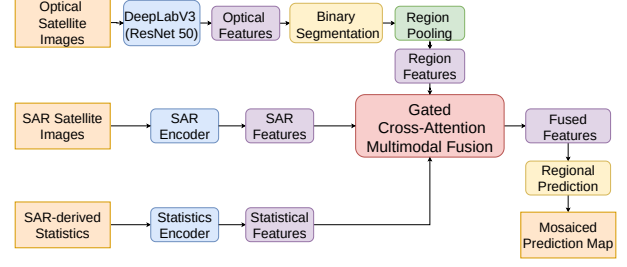


Fig. 3: Proposed multimodal fusion architecture.

be complemented by feature-rich optical imagery. Therefore, to improve robustness under challenging imaging and environmental conditions, we leverage the complementary properties of optical and SAR imagery and propose a multimodal fusion framework that combines optical features with SAR information using region-based representations. Sentinel-2 RGB bands and Sentinel-1 SAR images are spatially aligned and split into non-overlapping tiles. For each tile, the optical, SAR and SAR-derived statistics (Gaussian parameters in Eq. 2) are processed in parallel streams before being fused.

Each optical tile is encoded using DeepLabV3 with ResNet50 to produce a feature map. A lightweight binary segmentation head is used to distinguish between foreground and background. Each tile is then split into superpixel regions and region-level descriptors are obtained by pooling the optical features over each superpixel. SAR tiles are processed using a dedicated convolutional encoder to extract spatial representations from the SAR channels. In parallel, tile-level SAR statistics ($VH \mu$, $VV \mu$, $VH \sigma$, $VV \sigma$) are passed through a separate encoder which captures global characteristics that complement the spatial SAR features. Multimodal fusion combines region level image features and environmental indices using a gated cross-attention mechanism.

The dataset comprised 5,475 training and 1,449 test samples. Training used a region-level categorical cross-entropy loss, computed between the predicted and ground-truth class labels for each superpixel. To combat class imbalance, we calculate class weights based on the frequency of each class within the split and pass these weights into the loss function. An overview of the proposed architecture is shown in Fig. 3.

To quantify the contribution of each modality we compare four variations of the model: RGB-only using Sentinel-2 optical bands; RGB + SAR which fuses optical and SAR features via gated cross-attention; RGB + Stats which combines optical imagery with SAR-derived statistical features; and finally an RGB + SAR + Stats which combines all modalities. Training masks were produced using a semi-automated process and the three classes (low, mid and high) were chosen as a proxy for algal bloom severity. All of the variations share the same backbone architecture, training procedure and evaluation protocols to ensure a fair comparison. Models were trained using the Adam optimizer, with separate learning rates for the back-

bone and task-specific heads. Hyperparameters were tuned where required to accommodate differences in input modalities. Models were trained for up to 50 epochs on five independently resampled datasets using random seeds, with early stopping based on validation performance. Additional experiments were performed using super-resolution Sentinel-2 imagery generated using the Hybrid Attention Transformer (HAT) method by Chen et al.[16]. During training, a hybrid sampling strategy was used in which regular and super resolution image tiles were sampled with equal probability.

3. RESULTS AND DISCUSSION

3.1. Analysis of statistical modelling with SAR

We cross-validated each available Sentinel-1 acquisition against optical RGB imagery acquired within a ± 2 -day temporal window, to ensure VV backscatter patterns primarily reflect algal activity rather than wind-driven effects, we selected the *lowest possible* wind-speed SAR acquisition available for each month across the 2023–2025 period, using daily mean wind speed reanalysis data to identify the calmest conditions. Although a conservative threshold of 2 m/s was established as the ideal limit for HAB interpretation, several months had no acquisitions below this threshold. Consequently, the lowest available wind-speed images were retained, even when exceeding 2 m/s, to maintain monthly temporal continuity.

Fig. 4 reveals a distinct seasonal cycle in the low-wind acquisitions: winter months (November–March) exhibit consistently low and spatially uniform VV backscatter, while late spring through autumn (April–October) display elevated and spatially heterogeneous patterns, coinciding with increased algal and aquatic vegetation activity. The following acquisitions exceeded the 2 m/s threshold but represent the calmest conditions for their respective months: 17 Jan 2023 (3.27 m/s), 6 Apr 2023 (3.21 m/s), 14 Jul 2023 (3.55 m/s), 17 Dec 2023 (6.39 m/s), 5 Mar 2024 (4.25 m/s), 26 Nov 2024 (2.30 m/s), 30 Jan 2025 (5.37 m/s), and 28 Feb 2025 (3.13 m/s). On these dates, elevated wind-driven surface roughness produces high VV backscatter (appearing as reddish tones) that does not follow the expected seasonal trend and should not be attributed to algal activity.

3.2. Multimodal fusion analysis with ablation study

Table 1 compares four configurations: (1) RGB, (2) RGB + SAR, (3) RGB + statistical features and (4) RGB + SAR + statistical features (the proposed architecture). All models were trained under identical settings to ensure a fair comparison, with performance differences primarily attributable to the included modalities. Example outputs are shown in Fig. 5.

The RGB-only model shows strong performance on low- and background classes; however, its mid- and high-performance are weaker. Indicating reduced class separation. Incorporating SAR features improves low- and mid-

density regions, although it has lower overall performance. Adding statistical features with the RGB images improves mid-density performance (0.135 vs 0.108). The strongest overall configuration is the RGB + SAR + Stats model which combines the complementary strengths of each modality to achieve higher overall mIoU (0.348 ± 0.022) and an increased macro F1 score (0.470) indicating more balanced performance across all classes. Using super-resolution for augmentation within the training process made the differences between low and mid classes more prominent, improving the class separation and giving an mIoU of 0.359 ± 0.038 .

As no directly comparable benchmark methods exist for this specific task, approaches from related HAB monitoring studies were selected to provide contextual performance comparisons, rather than direct state-of-the-art benchmarking. The two comparison approaches included in Table 1 are therefore implemented as representative reference baselines that have been adapted to the same dataset and pipeline, rather than a strict reproduction of the original methods. This provides contextual performance references while evaluating the contribution of SAR and SAR derived statistics within the proposed method. Jeong et al. [10] uses a shallow custom CNN architecture, while Phetanan et al. [11] use a Faster R-CNN based architecture. All variations of our proposed method outperform both reference baseline approaches across most metrics. Overall, these results indicate that augmenting optical imagery with SAR data and derived statistical descriptors enhances robustness and consistency of algal bloom predictions, with relative gains of up to 125.8% overall mIoU and 40.4% macro-F1 score over baseline reference methods.

4. CONCLUSION

This study demonstrates the viability of a multimodal framework that integrates Sentinel-2 optical imagery with Sentinel-1 SAR data for detecting and monitoring algae spread in Lough Neagh. The proposed method is twofold: 1) statistical analysis on the SAR data that captures a holistic understanding of the algae spread, and 2) a feature-rich processing pipeline that fuses the complementary information from both modalities (SAR and optical). Analysis confirms that the SAR VV polarisation channel serves as a robust proxy for bloom events, characterised by a marked +6 dB backscatter enhancement and a distinct statistical shift toward multimodal distributions in nearshore zones. The multimodal approach achieves more robust and consistent performance than any single data source alone, particularly under the challenging conditions of persistent cloud cover and heterogeneous near-shore environments. Future work will explore incorporation of hyperspectral and related data products, alongside an expanded dataset including multiple lakes, to support the development of more capable multimodal models with improved feature extraction and generalisation, particularly in complex near-shore environments.

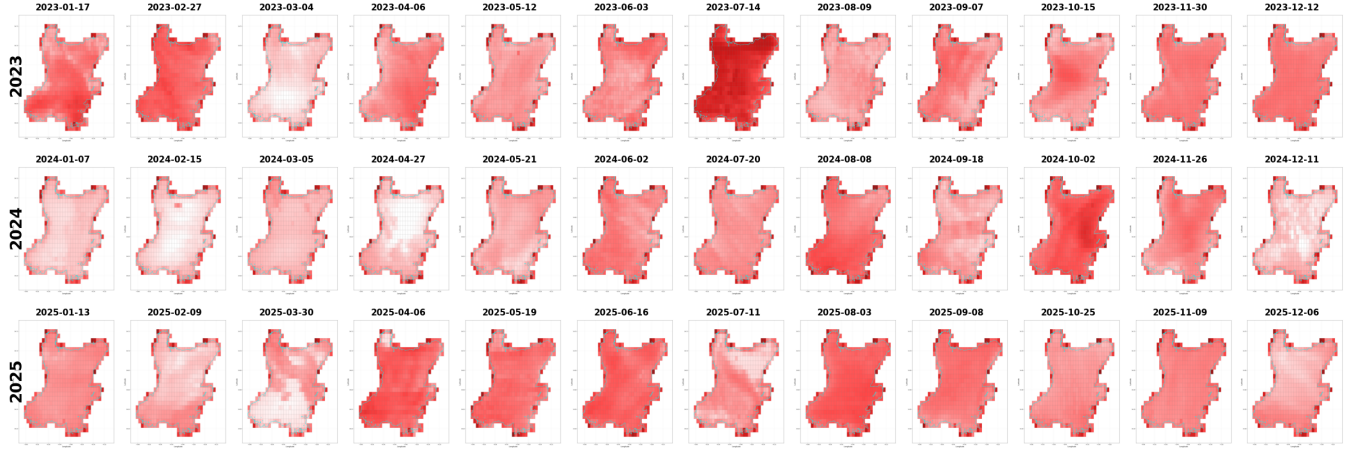


Fig. 4: SAR VV backscatter for three consecutive years (2023-2025) with standardised intensity scale (-26.8 to -9.5 dB), providing a holistic snapshot of the bloom infestation. White and dark red represent bloom-free and infested water surfaces, respectively. Columns represent selected dates in each month (January-December).

Table 1: Region-level prediction results for multimodal ablations and representative reference baselines.

Configuration	mIoU (All)	mIoU (fg)	mIoU (low)	mIoU (mid)	mIoU (high)	mIoU (bg)	macro F1
RGB	0.335 ± 0.019	0.225	0.461	0.108	0.106	0.663	0.453 ± 0.020
RGB + SAR	0.337 ± 0.018	0.233	0.477	0.118	0.103	0.648	0.457 ± 0.019
RGB + Stats	0.341 ± 0.008	0.238	0.477	0.135	0.101	0.651	0.462 ± 0.022
RGB + SAR + Stats	0.348 ± 0.022	0.243	0.483	0.135	0.110	0.663	0.470 ± 0.029
↪ + Super Resolution	0.359 ± 0.038	0.252	0.478	0.170	0.109	0.681	0.483 ± 0.051
Jeong <i>et al.</i> '25 [10]	0.159	0.211	0.282	0.123	0.228	0.004	0.344
Phetnan <i>et al.</i> '25 [11]	0.167	0.141	0.031	0.096	0.145	0.141	0.307

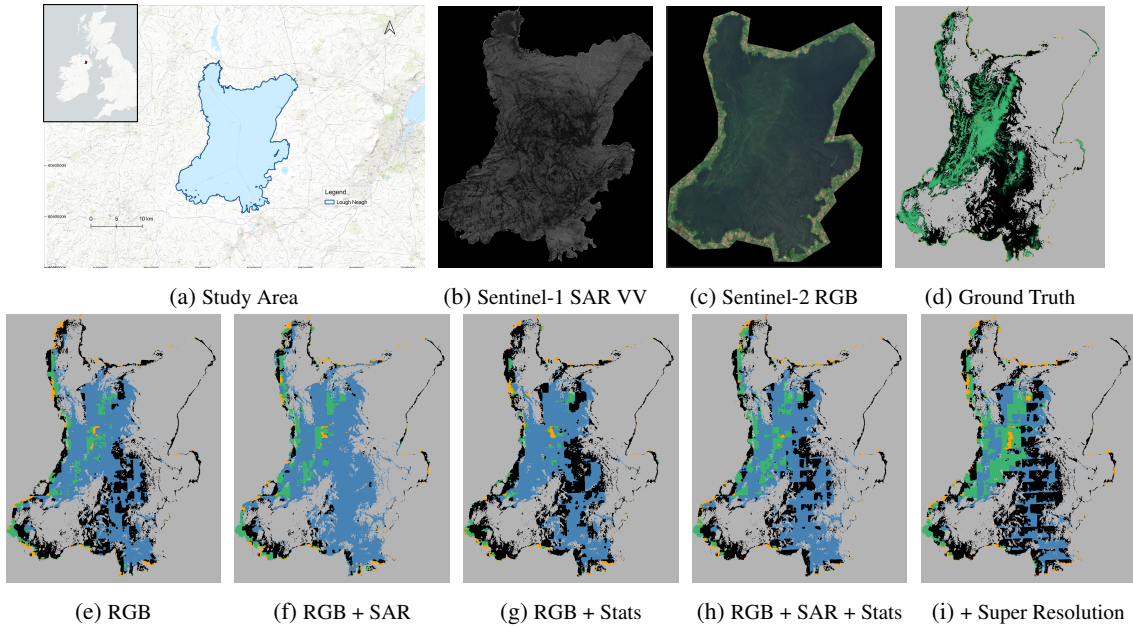


Fig. 5: Comparison of inputs, ground truth and prediction maps for algae bloom severity. (blue: low, green: mid, orange: high).

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